

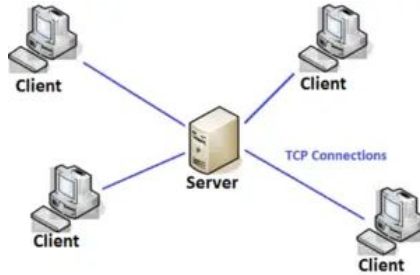
Cloud Computing

Today's session

- Section 1
 - Historical / Evolutionary perspective
 - Understanding the “Cloud” and what “Cloud Computing” means.
 - What makes the Cloud Computing advantageous
 - Analogies from the real world
- Section 2
 - Deployment Models (How are cloud services deployed in enterprises)
 - Service Models (How are cloud services served to consumers)
- Section 3
 - Cloud essentials
 - Relevance of standards for cloud deployments
- Summary
- Future of Cloud Computing

Section 1

The Evolution of Computing..



Mainframes

- 1970 - 80
- Large machines with central processing
- Dumb terminals to connect and use
- Very expensive

Client Server

- 1990 - 2000
- Distributed computing
- Smart clients with some local processing
- Remote connections
- Over the internet

On-prem server clusters

- 2000 - Present
- Enterprise owned computing resources
- Virtualization
- Automated deployments
- Web applications
- Remote clients (laptop)

Cloud

- 2010 - Present
- Scalable
- Global presence
- Low upfront cost
- Mobile clients
- X-aaS

What is Cloud computing



What is Cloud?



Cloud

Vast amounts of computing resources (CPU, RAM, Disk, Network) and software services available through the Internet in data centers all over the world.

Cloud computing

Technology and processes used to store, manage and process the data in the cloud.

Main advantages

- Instant, unlimited scalability (elastic)
- Access through Internet (from anywhere)
- Cost effective, measured service
- Global presence
- Fault tolerant, High availability
- Economies of scale
- **Lets you focus on what you do best (and forget the rest)**

Q1. Scalability

Why Elastic ?



Grow as your business scales (up or down).



Q2. Global Presence

Why Global ?



Global Citizen Usage Readiness
Data residency regulations and laws
Disaster readiness.



Q3. Economies of scale



What does that mean ?

- Supplier side economics
 - Standardization of hardware and therefore large scale bulk buying
 - Power cost savings (choice of datacenter locations)
 - Automation for maintenance/repair for millions of instances
- Consumer side economics
 - Demand aggregation
 - Pooling as a solution Time dependent traffic. (lean and peak periods)
 - Addressing unpredictable growth



Game changer

Huge computing resources create new possibilities & innovation opportunities

Can you name some innovative services we use today, that did not exist around 15 years ago?

- IoT, Home automation (Alexa, Google Home)
- Health tracking (Fitbit)
- Navigation with real time traffic data (Google Maps)
- Video on demand (Netflix)
- New communication modes (Twitter, WhatsApp)
- Unlimited storage (Dropbox, Google Photos)
- E-Commerce (Amazon, Flipkart) and Instant Delivery (Zomato, Blinkit..)



Paradigm Shifts

It is a fundamental shift in the way we engage and interact with the world and ultimately live our lives.

Cloud technology is changing how we work, study and fulfil our daily tasks.

An Analogy: Eating Dinner @ Home

On-premise meal

- Dal, Rice, Flour
- Vegetables
- Spices, Oil
- Water
- Stove, Gas
- Recipe
- Chef
- Table, Cutlery



Cloud Meal

- Select the food to eat
- Pay
- Wait for delivery
- Enjoy :)



More analogies of paradigm shifts from some of you ?



Hint :

1 - Music

2 - Use of public infrastructure

Many examples from the public domain

**Building your own house
on a plot of land**

vs

Buying/Renting apartments

**Buying a car to commute to
college**

vs

**Using metro or public
transport or car pool to
commute to college**



Section 2

Cloud Computing Deployment Models

Types of Cloud

Depending on your need and budget, there are multiple cloud deployment models.

Private Cloud

- Complete control and ownership
- Security
- Not suitable for small enterprises
- Private network/VPN
- Deep technical skills
- Government agencies, healthcare providers, banks, business that handle a lot of sensitive data

Hybrid Cloud

- Flexibility
- Useful when some information cannot be put in public cloud
- Complex
- Deep technical skills
- Companies that want to keep critical apps and data private – and still use public cloud services for non-critical use cases. *Criticality* here is to do with data classification from a security/compliance/regulatory perspective.

Public Cloud

- Run by third party
- Shared resources
- No upfront cost
- Super agile, elastic
- Accessed over internet
- Flexible cost based on consumption
- Take advantage of the latest apps and elastic infrastructure while keeping costs low

Some organizations may choose more than one option too - depending on the use cases



- ZDFC Bank Core Banking Department
 - Private Cloud
- ZDFC Bank Marketing Systems.
 - Public Cloud
- ZDFC Bank User Notifications on Mobile (SMS/WhatsApp/App Notification)
 - Hybrid Cloud

Service Models on the Cloud

Personas leveraging the Cloud



Even when choosing public cloud, there are multiple categories of services based on the users' needs

- Business users or consumers

- Software that just works for me e.g. Gmail, Dropbox, WhatsApp



- Developers

- Development platform, tools to build my apps and runtimes to deploy them



- System administrators

- Computing resources (Network, Storage, Virtual machines)



Categories of Cloud Services

Three categories of services typically provided by cloud providers

Infrastructure as a Service

- Computing resources
- On-demand infrastructure resources e.g. Compute, Storage, Networking
- User controls O/S, Middleware, apps, data
- Deploy and run arbitrary software
- Pay for the resources you use



Platform as a Service

- Development platform to build and deploy apps
- Users focus on application development
- Runtime, middleware, O/S, servers, storage, networking provided
- Pay for the services you use - resources and software



Software as a Service

- Software applications to use without local resources *or installation*
- Business users, consumers
- Focus on the tasks
- Everything managed by the provider - apps, data, middleware, O/S, computing resources



Categories of Dinner making



- Cutlery
- Table
- Drinks
- Gas, Stove
- Recipe
- Spices
- Vegetables
- Dal
- Rice

Cook at home



- Cutlery
- Table
- Drinks
- Gas, Stove
- Recipe
- Spices
- Vegetables
- Dal
- Rice

Ready to cook

- Cutlery
- Table
- Drinks
- Gas, Stove
- Recipe
- Spices
- Vegetables
- Dal
- Rice

Home delivery

- Cutlery
- Table
- Drinks
- Gas, Stove
- Recipe
- Spices
- Vegetables
- Dal
- Rice

Restaurant

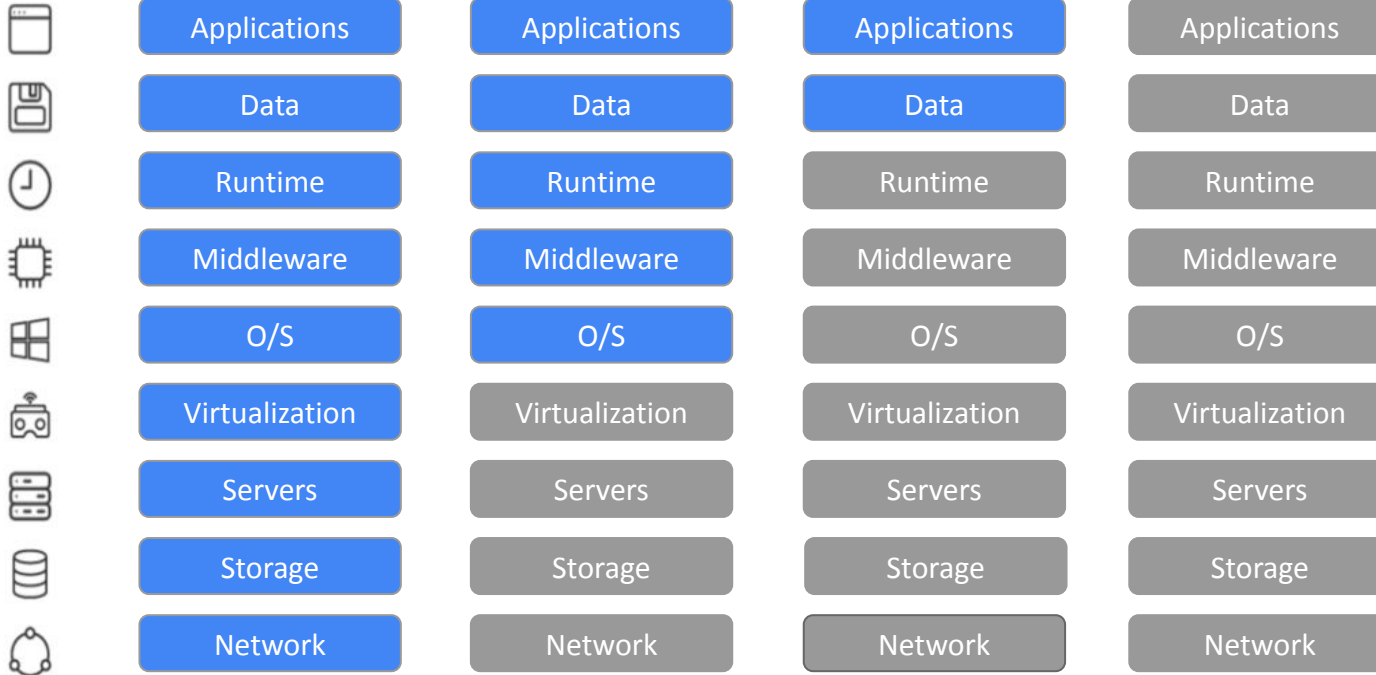
You bring

Provider gives

Categories of Cloud services

Consumer managed

Provider managed



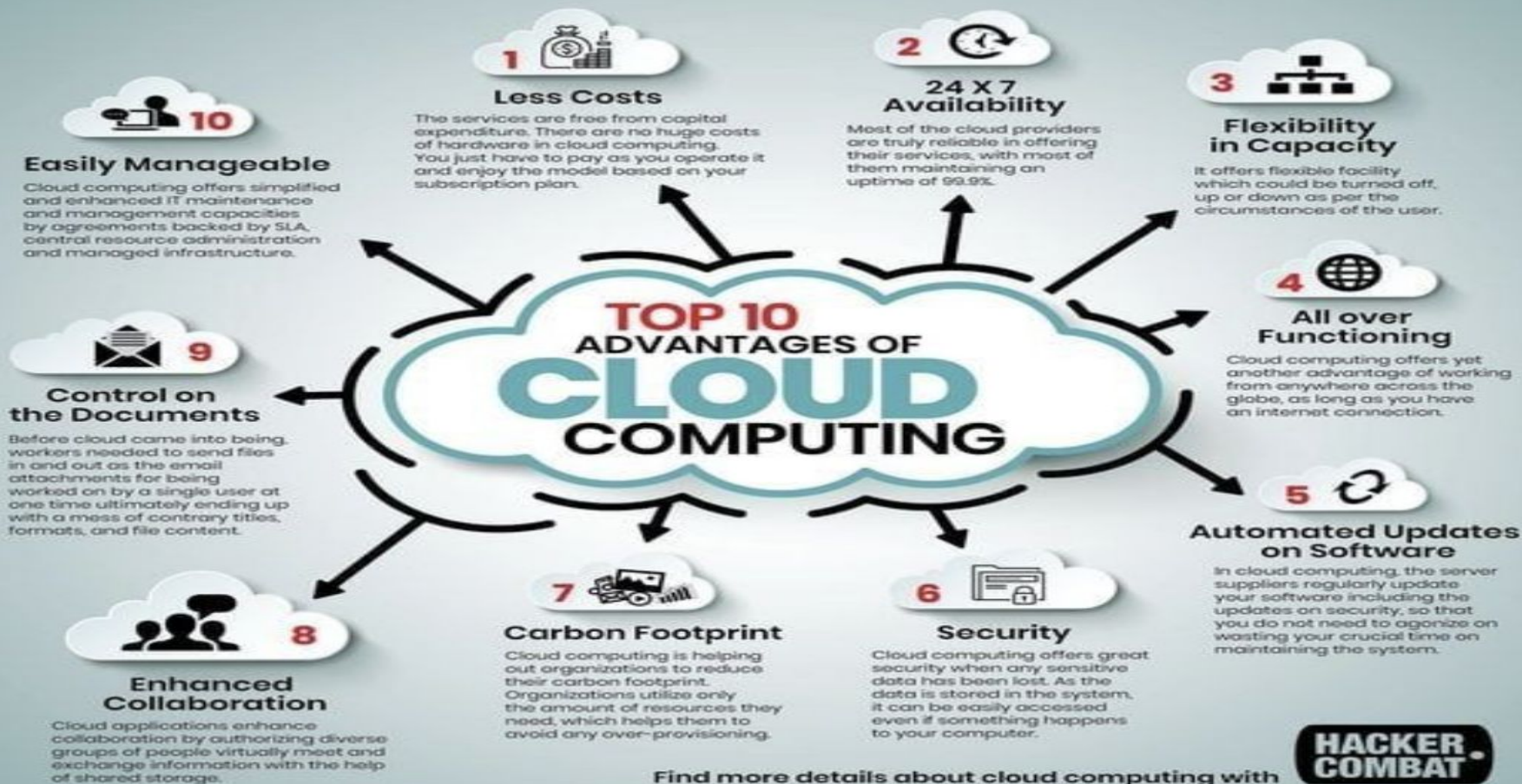
On-Prem

Infra-aaS
(IaaS)

Platform-aaS
(PaaS)

Software-aaS
(SaaS)

Advantages



Find more details about cloud computing with

**HACKER
COMBAT**

Real-World Applications

Web Hosting

Data Storage & Backups

Software application development and testing environments

Streaming Services

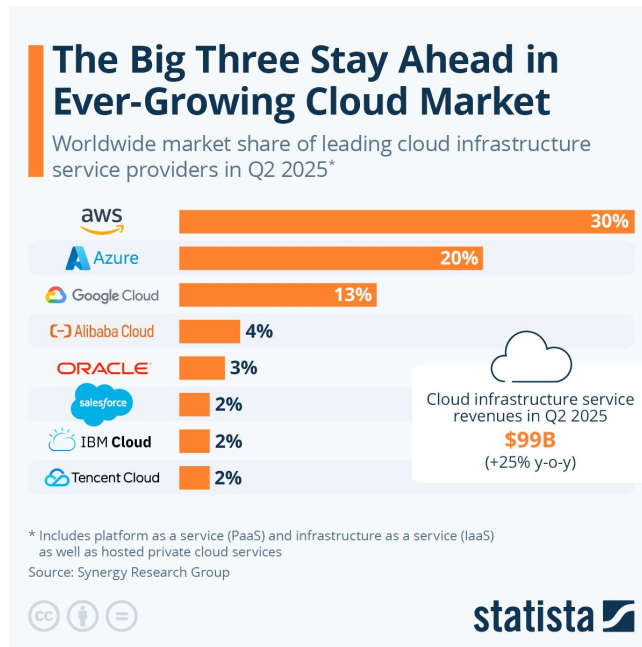
BigData Analytics

AI and ML workloads

Cloud Service Providers

Cloud Providers and their global penetration

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- IBM Cloud
- Oracle Cloud
- Salesforce Cloud
- Alibaba Cloud
- Linode



Key Capabilities provided by Providers

- On-demand self service to provision resources
- Pay-as you go pricing
- Broad network access
- Resource pooling
- Scalability and elasticity
- Security

Section 3

Cloud Essentials

Food for thought



What do you think the average size of an cloud deployment for a small business could be ?



What kind of challenges do you think arise ?

Security In the Cloud

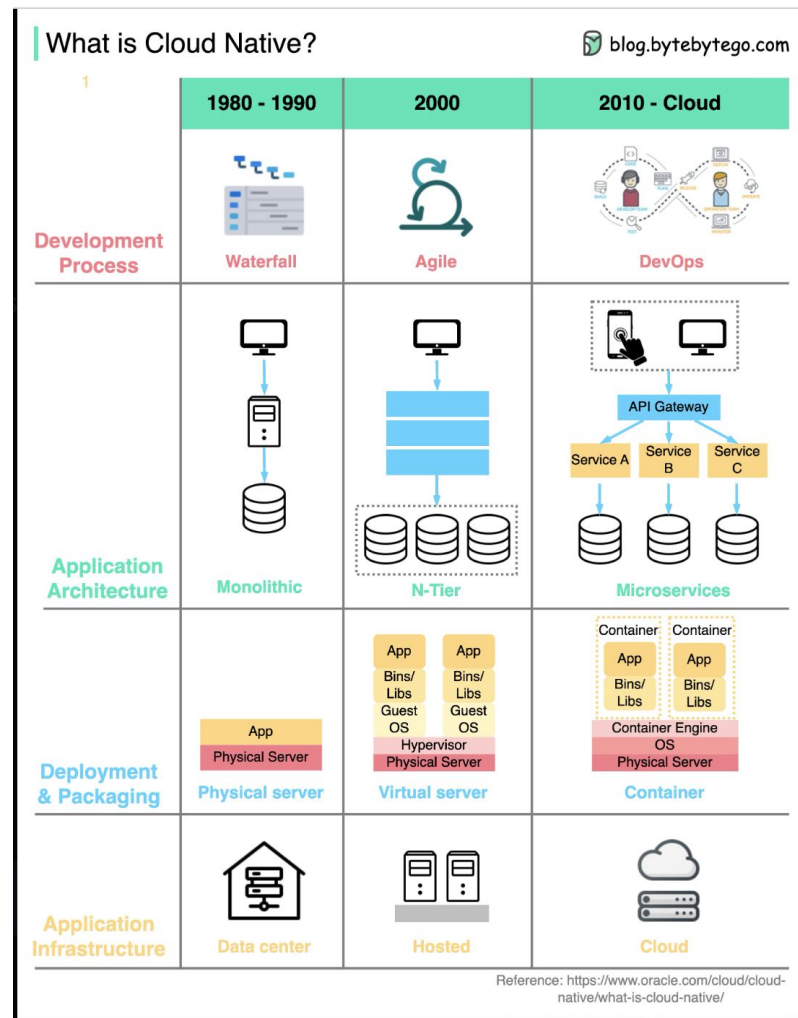
- Cloud providers are responsible for securing the infrastructure
- Customers are responsible for securing their data and applications
- Key security practices
 - Data encryption
 - Access control
 - Identity management.
- **Security is a shared responsibility**

Cloud scale operations

Very common to see public cloud deployments span tens of 1000s of compute nodes and 100s of applications.

Hard to maintain such complexity and keep services operational and reliable.

Cloud Native is a set of principles and processes to manage this complexity



SATKAAM

Adopting a cloud philosophy

Fundamental shift in how we manage software delivery life cycle (SDLC)

- Cloud native principles (12 factor app - see <https://12factor.net/>) to build and deliver applications.
- Platform Engineering principles to manage deployments
 - Everything operational “As code”
 - Infrastructure as code
 - Security with DevSecOps approach to build and deliver
- Operational Reliability through Site Reliability Engineering
 - monitors and improves the availability and performance of deployed software systems
- Continuous improvement culture
 - To manage costs, maintain operational reliability

Key technologies

Virtualization - Multiple virtual machines (VMs) on a single physical server through a “hypervisor” layer..

Containers - Lightweight, portable, and consistent runtime environments. Unlike VMs, containers share the same operating system kernel yet isolates the applications on it, making them more efficient in terms of performance and resource consumption.

Orchestration - Managing and scaling containers across a cluster.

Microservices - Breaking down applications into small, independent services

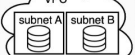
Service Mesh - Managing dependencies, communication and security between microservices



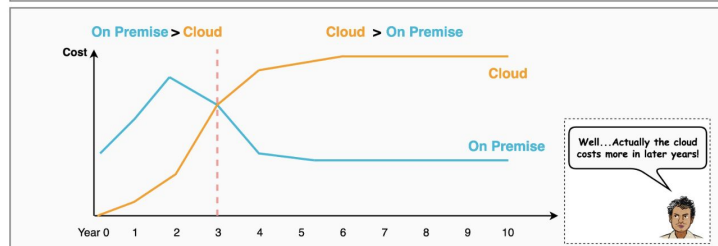
Is Cloud really cheaper ?



Does the Cloud Save Costs?

	On Premise	Amazon EC2	Amazon RDS
			
DBMS	App optimization	App optimization	App optimization
	Scaling	Scaling	Scaling
	High availability	High availability	High availability
	DB backups	DB backups	DB backups
	DB software patches	DB software patches	DB software patches
OS	DB software installations	DB software installations	DB software installations
	OS patches and config	OS patches and config	OS patches and config
Hardware	OS installation	OS installation	OS installation
	Server maintenance	Server maintenance	Server maintenance
	Rack and stack	Rack and stack	Rack and stack
	Power, network	Power, network	Power, network

Cloud saves so much trouble. It must have saved a lot of cost!



Source: Amazon doc, Cloud versus On-Premise Computing by Cameron Fisher

- Low starting costs for Cloud
- On-premise seems more efficient in long runs (>4 years)
- Benefits outweigh the cost differences though.
- Flexibility, horizontal scalability and reliability that comes with Cloud is immense.
- Can not be replicated easily in an on-premise deployment model without considerable efforts and costs.
- Generally organizations should opt for on-premise for only regulatory/compliance reasons

Summary

- Cloud Computing is about
 - vast amounts of computing resources (CPU, RAM, Disk, Network) and software services available through the Internet in data centers all over the world.
 - Instant, unlimited scalability (elastic)
 - Economies of scale
 - Lets you focus on what you do best (and forget the rest)
 - Standardized security and compliance
 - Different categories of services and deployment models
 - Changing the way we build and consume software
 - New practices around “cloud native” helping scale/manage this big shift

Certification Standards

Why Standards

- Building standardized safeguards for
 - Security
 - Privacy
- Regulatory requirements
 - Mandated by regulators
 - Sovereign governments
 - Industries
- Periodic assessments
 - Assessed by independent assessors / auditors for the certification (usually annually)
- Typically choose the standards and auditors after roadmap is

Standards

Standard	Area	Description
ISO / IEC 27001	Enterprise Information Security	Foundational international standard for a company-wide Information Security Management System (ISMS)
ISO / IEC 27017	Cloud Information Security	Information security guidelines for cloud environments.
ISO / IEC 27018	Personal Data Protection	Standards that help implement GDPR
NIST SP 800-53	Security & Privacy controls	Basis for federal information systems security
NIST SP 500-291	Cloud environment controls	Specific cloud requirements and interoperability & portability requirements for cloud computing environments
SOC2	Data protection	Audits the effectiveness of a service organization's controls related to security, availability, confidentiality, processing integrity, and privacy of data based on the AICPA's Trust Services Criteria
Cloud Security Alliance (STAR)	IAM, Compliance, Risk Mgt Framework Implementation	SOC2 + Controls matrix for cloud security

The future..

The future of cloud computing

Serverless Computing - Code not infrastructure

Edge Computing - Process data close to the source

AI/ML Services - Rapid improvements and expansion to model catalogs and AI based services available to deploy and use.

Quantum Services - Access to quantum computing resources (**Quantum as a Service**)

MultiCloud- Cloud agnostic deployments and solutions to more cheaper and secure workload management spanning a combination of cloud platforms.

Sovereign clouds - government regulations around in-country data residency of data and day-to-day operational management by sovereign national citizens.

Zero Trust Security - Security based on zero trust principles that can be extended to any cloud deployment models and vendor.



Thanks..

Addendum

What kind of clouds do you think the following organizations would make use of.

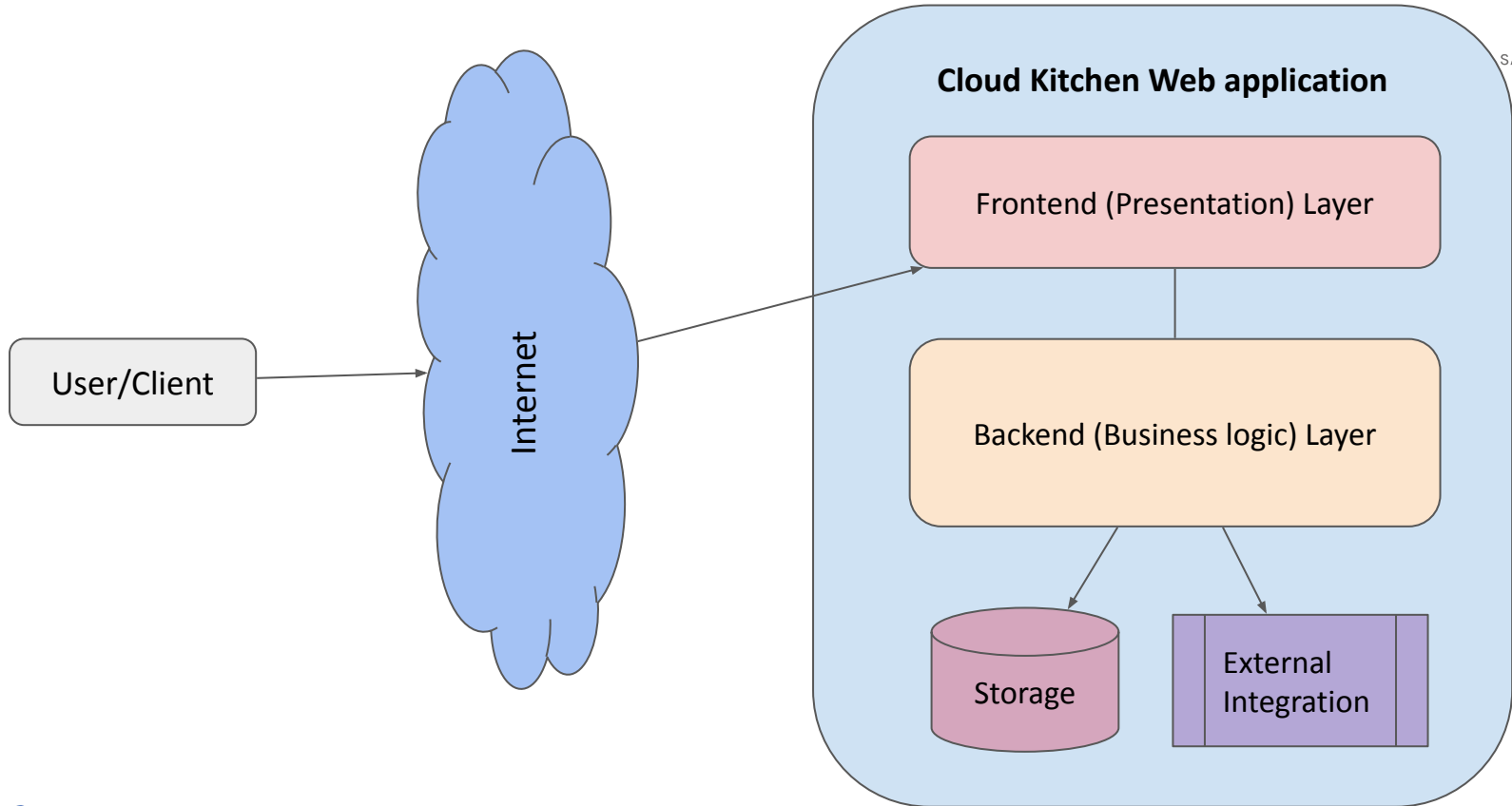


- CricInfo
 - Public Cloud
- Times Of India
 - Public Cloud
- RBI
 - Private Cloud
- UIDAI/Aadhar
 - Public Cloud. Right ?
 - No, Public Cloud Is Not Correct. It is Private Cloud
- Airtel
 - Hybrid Cloud

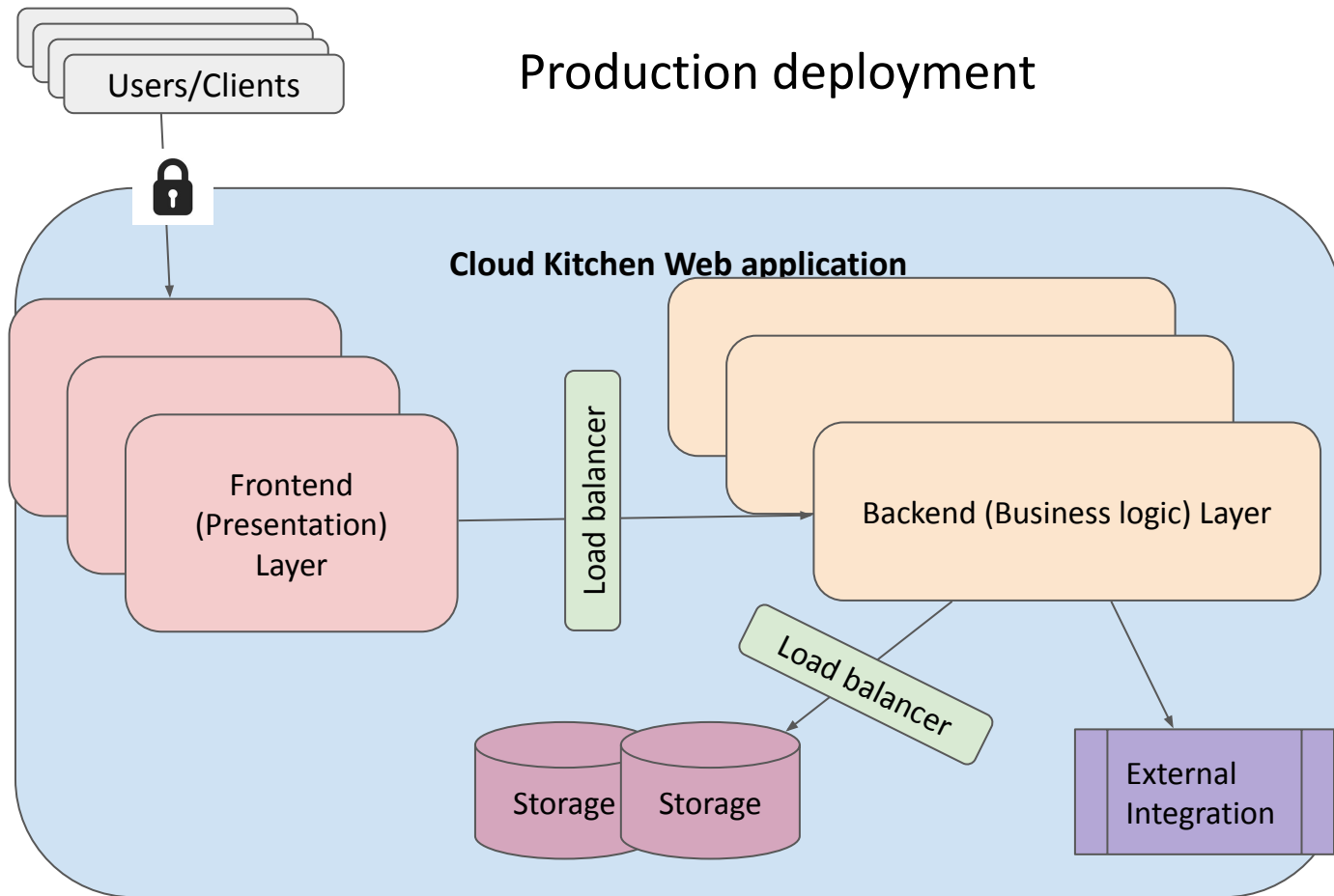


Typical structure of a cloud deployment

A typical Web application



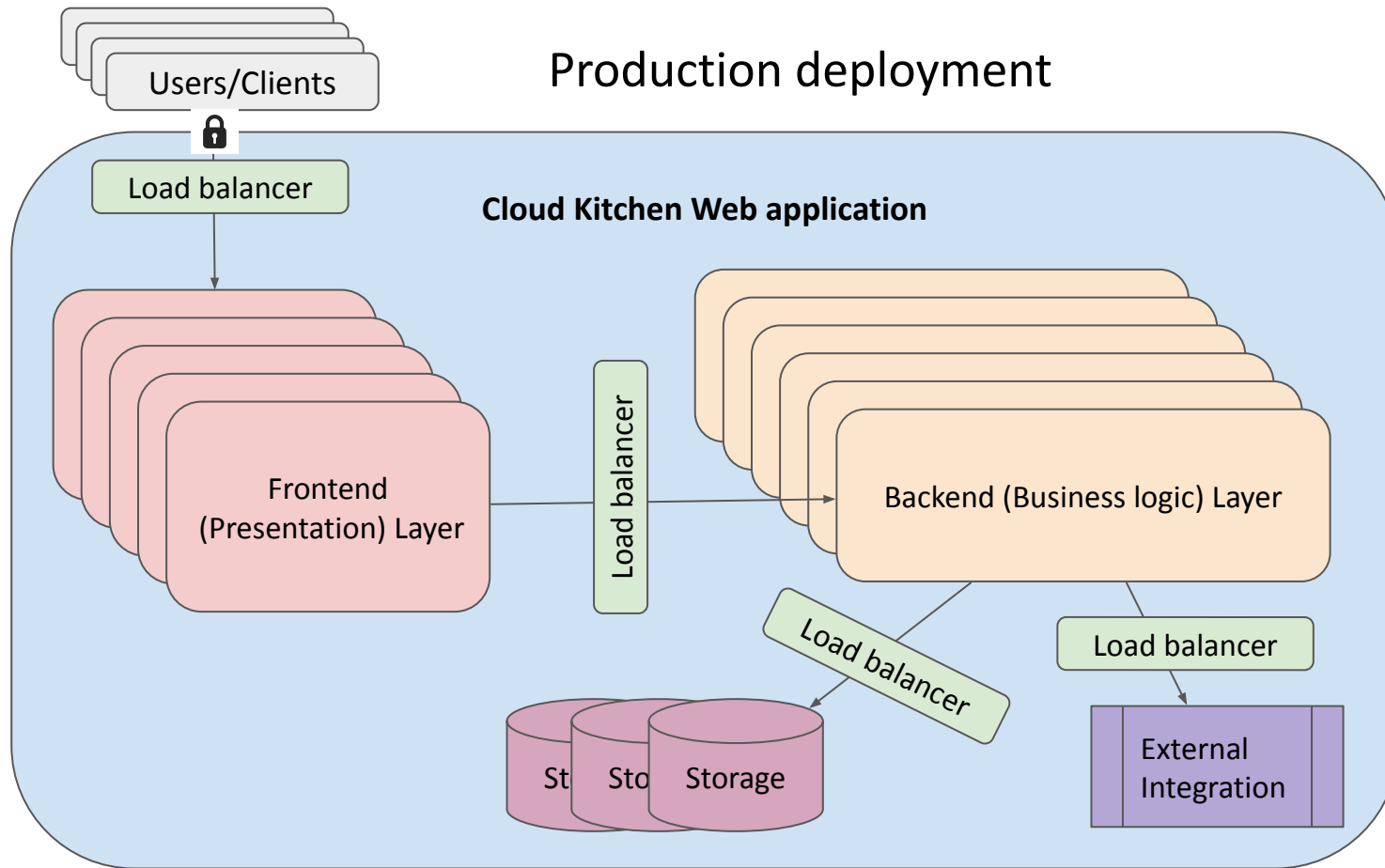
Production deployment



Redundancy

Fault Tolerance

Production deployment



Routing

Monitoring &
Auto scaling

Production deployment

